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Artificial Intelligence

Modules covered under the course

1. Introduction to Artificial Intelligence
2. Current and Future Industry Trends
3. Technical Roadmap to Artificial Intelligence



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Introduction to Artificial Intelligence

Understanding the Technology Beyond the Hype



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Agenda

- AI – Introduction, what people think
- Define AI, what actually is AI
- Why AI matters today
- History, Turing test, Dartmouth Conference , ELIZA
- AI - The New Millennium
- Generative AI – chatgpt
- Scope of AI, risk & limitations
- Key take-aways, conclusion



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How people think about AI!





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What AI actually is!

$$\begin{aligned} V^*(s) &= \max_a Q^*(s, a) \\ &= \max_a E_{\pi^*} \{ R_t \mid s_t = s, a_t = a \} \\ &= \max_{a \in \mathcal{A}(s)} E_{\pi^*} \left\{ \sum_{k=0}^{\infty} \gamma^k r_{t+k+1} \mid s_t = s, a_t = a \right\} \\ &= \max_{a \in \mathcal{A}(s)} E_{\pi^*} \left\{ r_{t+1} + \gamma \sum_{k=0}^{\infty} \gamma^k r_{t+k+2} \mid s_t = s, a_t = a \right\} \\ &= \max_{a \in \mathcal{A}(s)} E \{ r_{t+1} + \gamma V^*(s_{t+1}) \mid s_t = s, a_t = a \} \quad (3.14) \\ &= \max_{a \in \mathcal{A}(s)} \sum_{s'} \mathcal{P}_{ss'}^a [\mathcal{R}_{ss'}^a + \gamma V^*(s')], \quad (3.15) \end{aligned}$$

$$\frac{w}{\|w\|} \cdot (x_2 - x_1) = \text{width} = \frac{2}{\|w\|}$$

$$w \cdot x_2 + b = 1$$

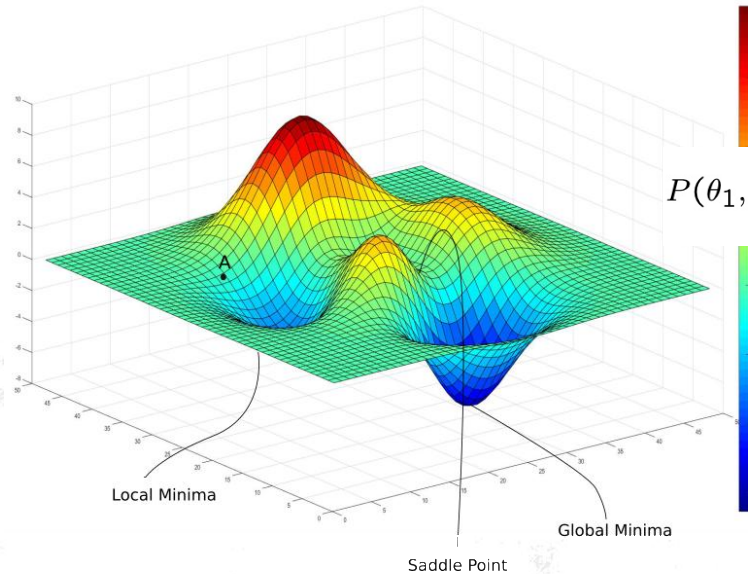
$$w \cdot x_1 + b = -1$$

$$w \cdot x_2 + b - w \cdot x_1 - b = 1 - (-1)$$

$$w \cdot x_2 - w \cdot x_1 = 2$$

$$\frac{w}{\|w\|} (x_2 - x_1) = \frac{2}{\|w\|}$$

2

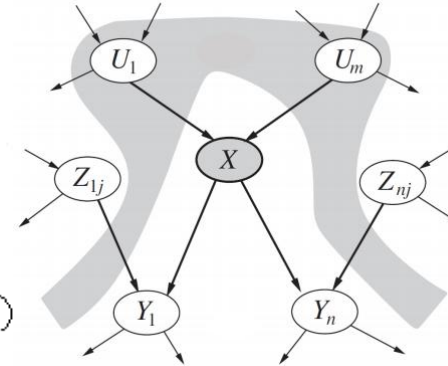


$$P(\theta_1, \theta_2, \dots, \theta_m) = \frac{\Gamma(\sum_k \alpha_k)}{\prod_k \Gamma(\alpha_k)} \prod_{k=1}^m \theta_k^{\alpha_k - 1}$$

$$L(\mu, \sigma^2; x_1, \dots, x_n) = \prod_{j=1}^n f_X(x_j; \mu, \sigma^2)$$

$$= \prod_{j=1}^n (2\pi\sigma^2)^{-1/2} \exp\left(-\frac{1}{2} \frac{(x_j - \mu)^2}{\sigma^2}\right)$$

$$= (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{1}{2\sigma^2} \sum_{j=1}^n (x_j - \mu)^2\right)$$





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Let us define A.I.



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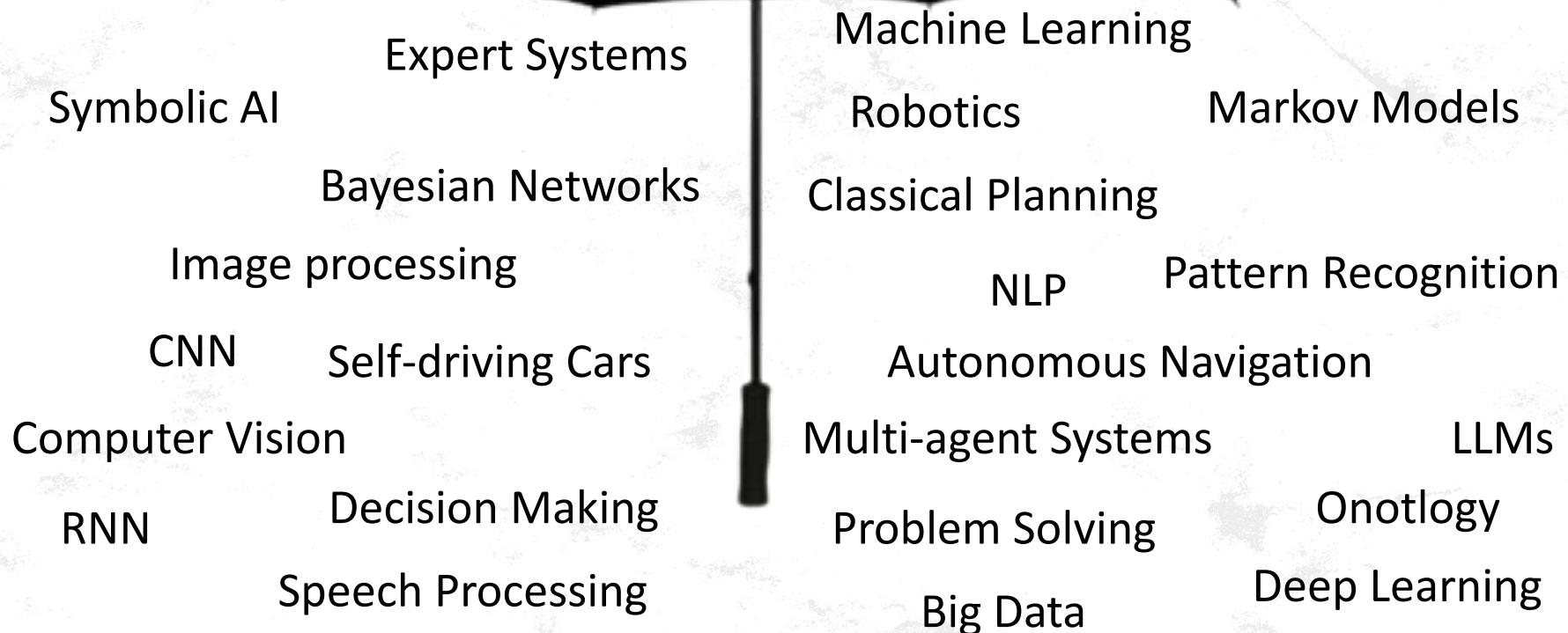




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A.I.





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Why AI Matters Today

- AI is transforming how work, services, and decisions are made:
 - It can handle scale beyond human capacity
 - It enables faster and more informed decisions
 - It improves consistency and efficiency
 - It can extract insights from complex data in limited time
- AI is becoming a **general-purpose technology**, like electricity or the internet.

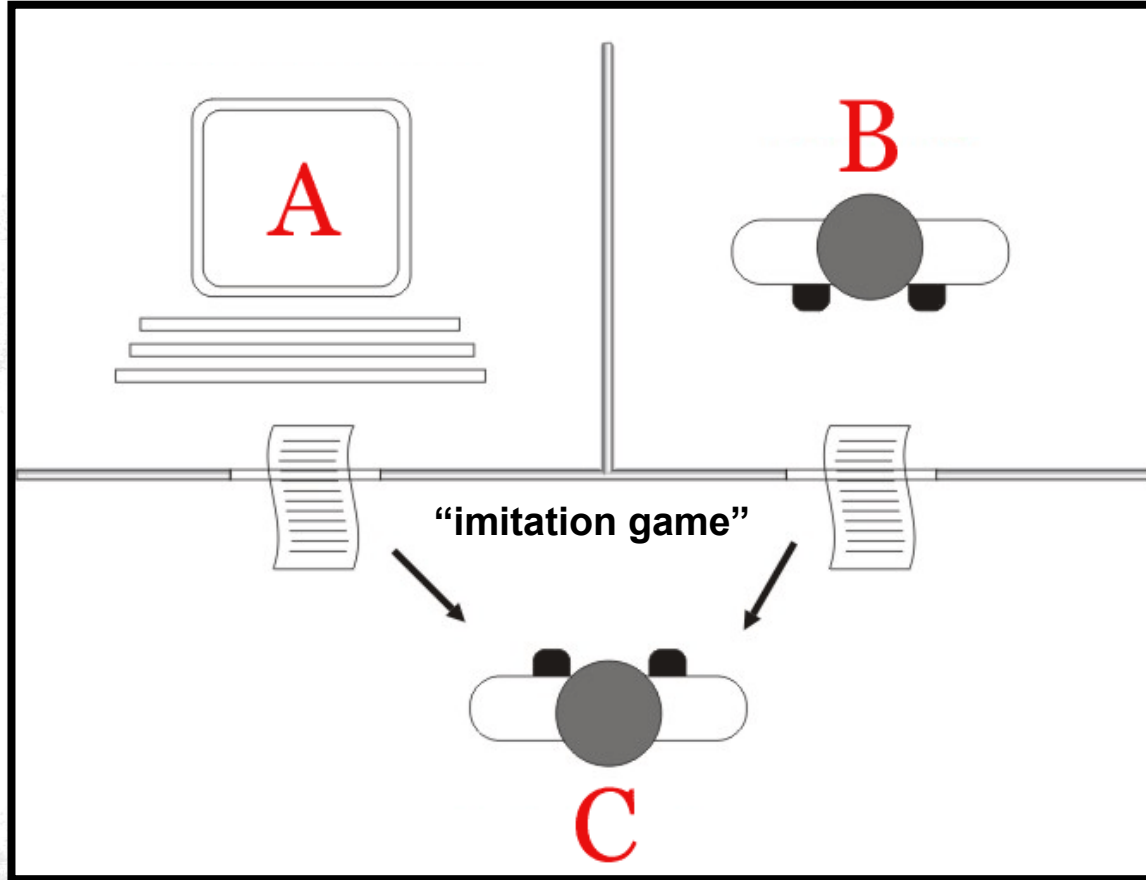


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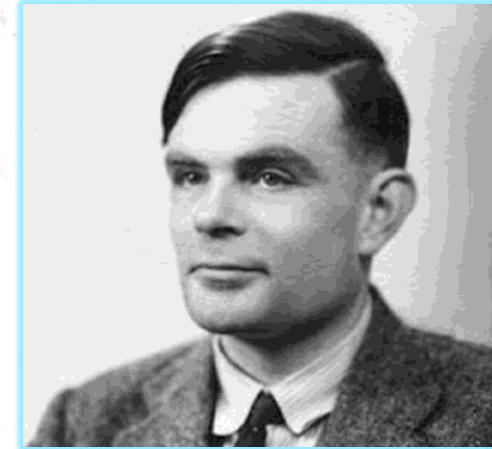
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The Turing test (1950)

- Developed by Alan Turing



https://en.wikipedia.org/wiki/File:Turing_test_diagram.png



The Loebner Prize

The Loebner Prize is an annual competition in artificial intelligence that awards prizes to the computer programs considered by the judges to be the most human-like. The format of the competition is that of a standard Turing test.



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Dartmouth Conference (1956)

- Dartmouth Summer Research Project on Artificial Intelligence organized by John McCarthy



From left:
Trenchard More, John McCarthy,
Marvin Minsky, Oliver Selfridge, and
Ray Solomonoff

<http://www.dartmouth.edu/~vox/0607/0724/ai50.html>

- Birth of the term “Artificial Intelligence (AI)”
- Gave AI, its mission, its first success and its major players



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The ELIZA Program (1964-66)

- Created at the MIT Artificial Intelligence Laboratory by Joseph Weizenbaum
- "ELIZA - A Computer Program for the Study of Natural Language Communication between Man and Machine," Communications of the Association for Computing Machinery 9 (1966): 36-45.

```
Welcome to

EEEEEE LL      IIII ZZZZZZZ AAAAA
EE      LL      II      ZZ  AA  AA
EEEEEE LL      II      ZZZ  AAAAAA
EE      LL      II      ZZ  AA  AA
EEEEEE LLLLLL IIII ZZZZZZZ AA  AA

Eliza is a mock Rogerian psychotherapist.
The original program was described by Joseph Weizenbaum in 1966.
This implementation by Norbert Landsteiner 2005.

ELIZA: Is something troubling you ?
YOU:   Men are all alike.
ELIZA: What is the connection, do you suppose ?
YOU:   They're always bugging us about something or other.
ELIZA: Can you think of a specific example ?
YOU:   Well, my boyfriend made me come here.
ELIZA: Is it important to you that your boyfriend made you come here ?
YOU:   He says I'm depressed much of the time.
ELIZA: I am sorry to hear that you are depressed.
YOU:   It's true. I am unhappy.
ELIZA: Can you explain what made you unhappy ?
YOU:   █
```

https://en.wikipedia.org/wiki/ELIZA#/media/File:ELIZA_conversation.jpg



https://upload.wikimedia.org/wikipedia/commons/1/17/Joseph_Weizenbaum.jpg



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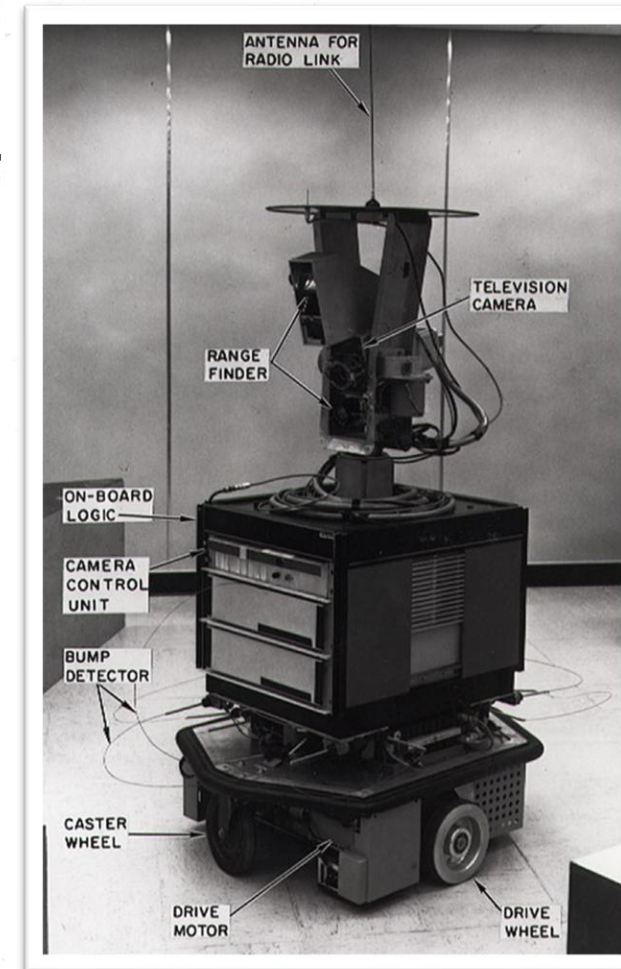
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Shakey the robot (1966-72)

- First general-purpose mobile robot to be able to reason about its own actions
- Combined research in robotics, computer vision, and natural language processing



https://en.wikipedia.org/wiki/Shakey_the_robot#/media/File:IEEE_Shakey_Team_at_dedication_of_IEEE_plaque_commemorating_their_achievement.jpg



https://en.wikipedia.org/wiki/Shakey_the_robot#/media/File:SRI_Shakey_with_callouts.jpg



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The First AI Winter(1974–1980)

- Reasons of failures:
 - Over enthusiasm and very high claims as what AI can do
 - Limited Computing capability at that time
 - Only “toy” AI programs could be implemented
 - Limited availability of knowledge base
 - Shortage of funding





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The Short Revival (1980–1987)

- Rise of Expert System in the industries (knowledge based systems)
- Japanese fifth generation computer project
- Revival of Connectionism
 - Discovery of back-propagation in Neural Networks
 - **Hopfield Networks**





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The Second AI Winter (1987–1996)

- The Expert systems couldn't keep the pace with industry demand and expansion
- The personal computers started flourishing and focus of industry shifted from AI to making compact computers
- The funding agencies started focusing on more pragmatic projects such as the Internet

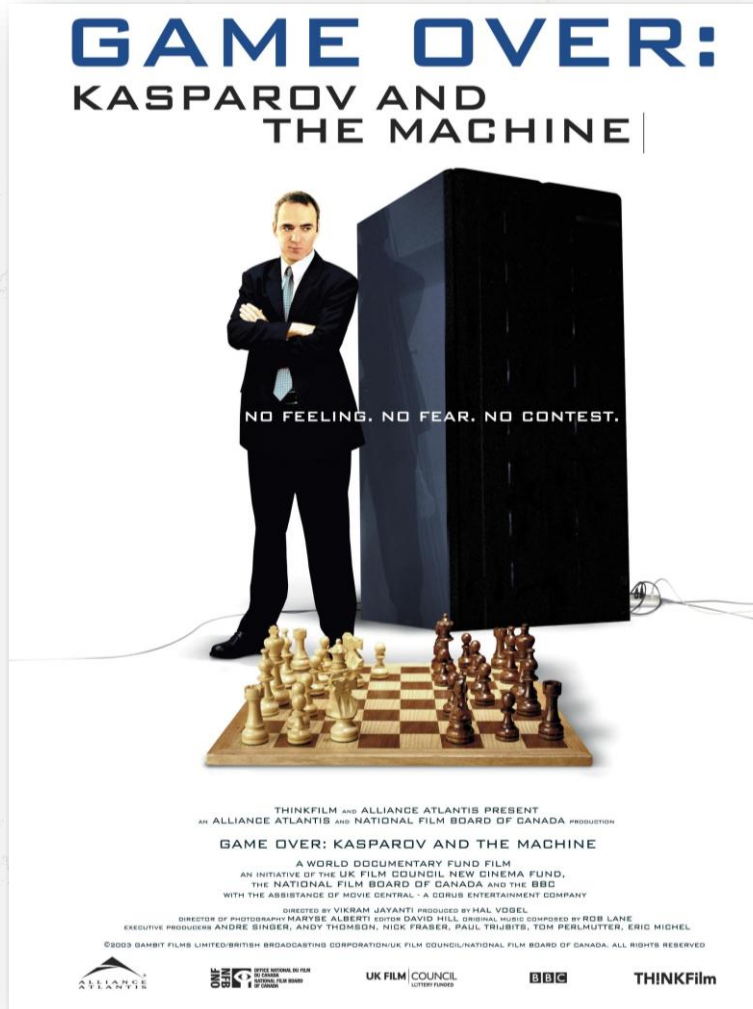




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The IBM Deep-Blue (1997)



<https://www.youtube.com/watch?v=NJarxpYyoFI>



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The New Millennium





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The New Millennium

- The DARPA Grand Challenge
 - Autonomous Driving



Stanley by Stanford University
Won the DARPA Grand Challenge **2005**



Winner of **2007** Urban Challenge,
Designed by CMU



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The New Millennium

IBM Watson wins the Jeopardy game against greatest champions (2011)





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The New Millennium

The era of drones



https://www.ted.com/talks/vijay_kumar_robots_that_fly_and_cooperate?language=en#t-926912



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The New Millennium

In 2016, Google's DeepMind Project AlphaGo defeats world's Go Champion Lee Sedol in the board game

19×19 board: $\sim 2.082 \times 10^{170}$ (i.e., a 2 followed by 170 zeroes)





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The New Millennium

- In October 2017, Sophia became the first robot to receive citizenship of any country.
- In November 2017, Sophia was named the United Nations Development Programme's first ever Innovation Champion, and is the first non-human to be given any United Nations title.





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2019 Turing Award Won by 3 Pioneers in Artificial Intelligence



From left, Yann LeCun, Geoffrey Hinton and Yoshua Bengio. The researchers worked on key developments for neural networks, which are reshaping how computer systems are built.



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ChatGPT



- **ChatGPT** (Chat Generative Pre-Trained Transformer)*
 - A large language model (LLM)-based chatbot developed by **OpenAI**
 - Launched on **November 30, 2022**,
 - Notable for enabling users to refine and steer a conversation towards a desired length, format, style, level of detail, and language used.
 - Successive prompts and replies, known as **prompt engineering**, are taken into account at each stage of the conversation as a context



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2024 Nobel Prize in Physics



NOBELPRISET I FYSIK 2024
THE NOBEL PRIZE IN PHYSICS 2024



KUNGL.
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THE ROYAL SWEDISH ACADEMY OF SCIENCES



Photo: Mary Weikum

John J. Hopfield
Princeton University, NJ, USA



Photo: University of Toronto

Geoffrey E. Hinton
University of Toronto, Canada

"för grundläggande upptäckter och uppfinningar som möjliggör maskininlärning med artificiella neuronnätverk"

"for foundational discoveries and inventions that enable machine learning with artificial neural networks"

THE
NOBEL
PRIZE



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The Broad Scope of AI

- AI spans multiple capability layers:
 - Perception – vision, speech, sensing
 - Prediction – forecasting, risk analysis
 - Reasoning – planning and optimization
 - **Generation** – text, images, code
 - Action – robotics and control systems
- Chatbots represent only a small visible part of this ecosystem.

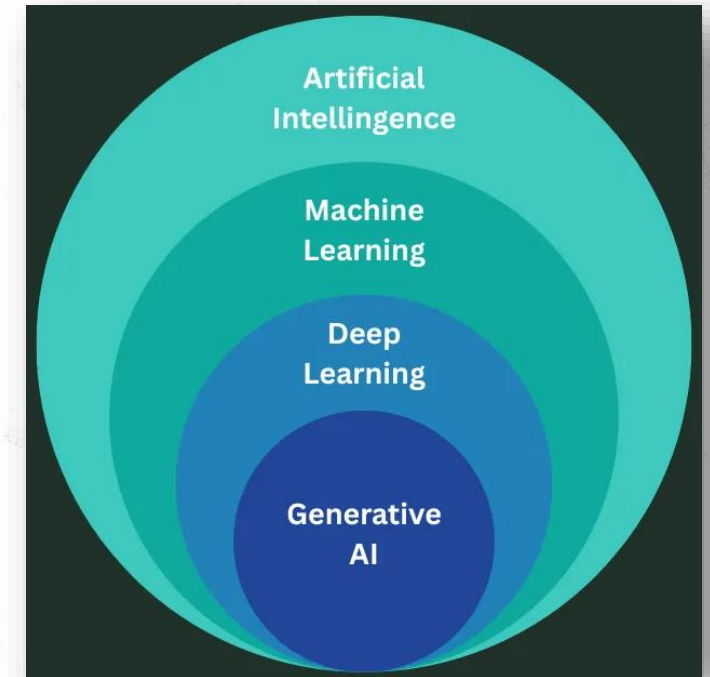


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Generative AI

- Generative AI (such as ChatGPT, Gemini, etc) focuses on creating content:
 - Text, images, audio, video, code
 - Natural language interaction
- Generative AI is:
 - **A subset of AI**
 - Powerful for communication and creativity
 - Not suitable for all tasks
- It complements—not replaces—other AI systems.





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Risks and Limitations of AI

- Across all AI systems, key risks include:
 - Bias from historical data
 - Lack of explainability
 - Errors and uncertainty
 - Over-automation
 - Privacy and security concerns
 - Hallucinations





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The Near Future of AI

- Likely developments:
 - AI embedded in most digital tools
 - Greater autonomy in systems
 - Increased regulation and standards
 - More private and on-device AI
 - Stronger integration with robotics
- AI will become less visible but more powerful.





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Key Take-Aways

- AI is:
 - Broad
 - Transformative
 - It is going to stay
- But:
 - Humans define goals
 - Humans set values
 - Humans remain responsible
- **AI is a tool for progress — not a replacement for human judgment.**



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Conclusions

- AI is transforming industries now
- Future AI is multimodal, autonomous, and domain-specific
- Adoption requires responsibility and strategic planning



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For Further Learning

- For additional skilling, advanced training in Artificial Intelligence and allied domains, please reach out to NIELIT.
- Visit **www.nielit.gov.in** or contact your nearest NIELIT Centre for guidance and support.



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धन्यवाद



THANK YOU



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